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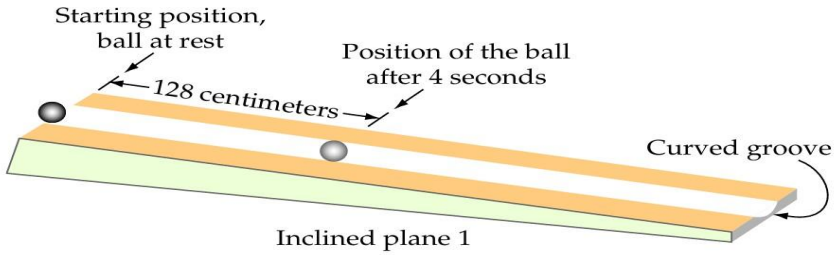
Math 111 Mathematics in the Modern World

Problem Set 1: Problem Solving

Instructions: Show your complete and comprehensive solutions in each of the following items. Use short bond paper. Minimize erasures. Write legibly.

1. (10 Points) Galileo (1564-1642) wanted to determine how the speed of a falling object changes as it falls. He conducted many free-fall experiments, but he found the motion of a falling object difficult to analyze. Eventually, he came up with the idea that it would be easier to perform experiments with a ball that rolls down a gentle incline, because the speed of the ball would be slower than the speed of a falling object. One objective of Galileo's inclined plane experiments was to determine how the speed of a ball rolling down an inclined plane changes as it rolls.

Examine the inclined plane shown below and the time-distance data in the following table.



Elapsed time t, in seconds	Distance d, in centimeters (cm), a ball rolls in t seconds (s)	
	Inclined plane 1	Inclined plane 2
0	0	0.0
1	8	6.5
2	32	26.0
3	72	58.5
4	128	104.0
5	200	162.5

Determine the distance a ball rolls, on inclined plane 1, during each of the following time intervals, Hint: To determine the distance a ball rolls in the interval $t = 1$ to $t = 2$ seconds, find the distance it rolls in 2 seconds and from this distance subtract the distance it rolls in 1 second.

a. 1st second: $t = 0$ to $t = 1$ second
b. 2nd second: $t = 1$ to $t = 2$ seconds
c. 3rd second: $t = 2$ to $t = 3$ seconds
d. 4th second: $t = 3$ to $t = 4$ seconds
e. 5th second: $t = 4$ to $t = 5$ seconds

2. (10 Points) Galileo conjectures that as a ball rolls down an inclined plane, its speed is increasing, because in each successive 1-second time interval after the first 1-second time interval, it rolls 2 units farther that it did in the preceding time interval of equal duration. This idea that the speed of a ball rolling down an inclined plane increases uniformly as it rolls downward was contrary to the popular belief, held in the early 1600s, that a ball rolls down an inclined plane at a constant rate.

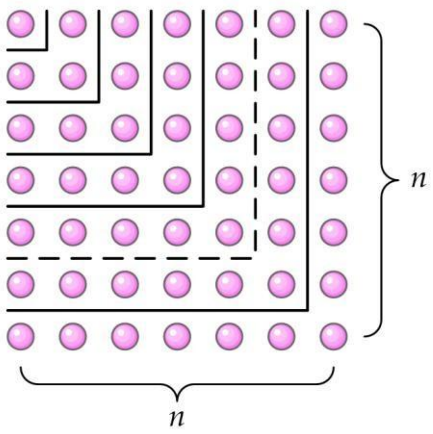
Galileo was also interested in the relationship between the amount of time a ball rolls and the distance it rolls.

Use inductive reasoning and the data in the inclined plane time-distance table, shown in item 1, to predict the answer to each question.

a. If the ball is allowed to roll on an inclined plane is doubled, what effect does this have on the distance the ball rolls?
b. If the time a ball is allowed to roll on an inclined plane is tripled, what effect does this have on the distance the ball rolls?
c. How far will a ball roll on inclined plane 1 in 6 seconds?
d. How far will a ball roll on inclined plane 1 in 1.5 seconds?

3. (10 Points) The ancient Greeks often discovered mathematical relationship by using geometric drawings. Study the accompanying drawing to determine what need to be put in place of the question mark to make the equation a true statement.

$$1 + 3 + 5 + 7 + \cdots + (2n - 1) = ?$$



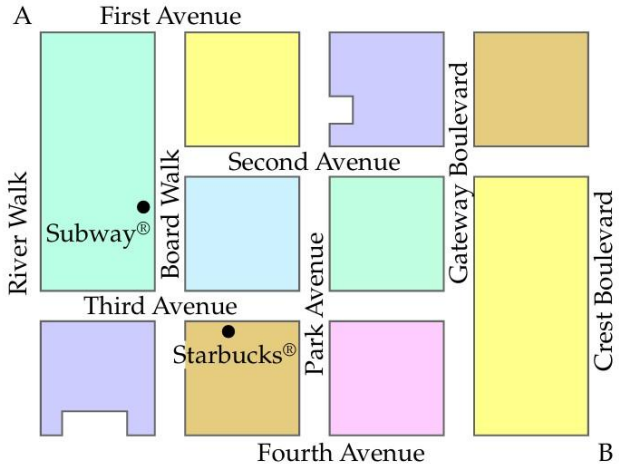
4. (10 Points) Make a conjecture for each of the following sums, where F_n represents the nth Fibonacci number. Present data as basis for your conjecture.

a. $F_n + 2F_{n+1} + F_{n+2} = ?$

b. $F_n + F_{n+1} + F_{n+3} = ?$

5. (10 Points) Use Polya's four-step problem solving strategy and the problem-solving procedures to solve the following problem.

Consider the following map. Tyler wishes to walk alone the streets from point A to point B. How many direct routes (no backtracking) can Tyler take?



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BSIS 1A

Deadline: February 26, 2026

Group 1 AGOT ALBAO ALBIT	Group 8 GEPULLANO GEPULLANO GULAYAN
Group 2 ALTERADO ASINO AURISTELA	Group 9 HINLAYAGAN JAVILLANO LAO
Group 3 BATIAO BELZA BENTONON	Group 10 LAYUPAN MACARINE MANAWAT
Group 4 BIJE BOLACOY BUCA	Group 11 MARISCAL MOLO MORALES
Group 5 CAGUD CALUNIA CANADA	Group 12 OMPOD OTTO PAGCAS PANTIA
Group 6 CEJAS CHAN DELAPAS	Group 13 PEÑAFIEL PIEDAD PROTASIO QUINIQUITO
Group 7 DOMINNO FLORES GARNICA	

BPA 1B

Deadline: March 2, 2026

Group 1 ALO ANDICO AZORES	Group 6 MIJARES MONTEDERAMOS NOPAL
Group 2 BAGO BENEMERITO CABALLERO	Group 7 ORTILANO PASAOL PIAÑAR
Group 3 CERIALES CONFESOR DIAZ	Group 8 QUIÑONES RAGO RIZON
Group 4 DUGHON ELCHICO JUNSAY	Group 9 SECUYA SUERTE TACLOBOS VILLAREN
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